

Biomedical Databases

GSND 5340Q, BMDA

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2025-06-02

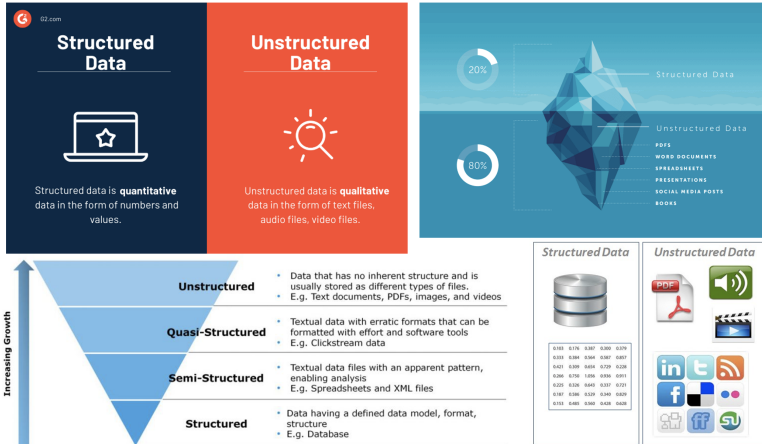
Biological Databases and resources

BIG DATA



Big Data has fundamentally changed how we look at science and business. Along with advances in analytic methods, they are providing unparalleled insights into our physical world and society

Structured vs. Unstructured data



Biology Is A Data Science

- ▶ Hundreds of thousand of species
- ▶ Million of articles in scientific literature
- ▶ Genetic Information
 - ▶ Gene names
 - ▶ Phenotype of mutants
 - ▶ Location of genes/mutations on chromosomes
 - ▶ Linkage (relationships between genes)

Data and Metadata

- ▶ Data are “concrete” objects
 - ▶ e.g. number, tweet, nucleotide sequence
- ▶ Metadata describes properties of data
 - ▶ e.g. object is a number, each tweet has an author
- ▶ Database structure may contain metadata
 - ▶ Type of object (integer, float, string, etc)
 - ▶ Size of object (strings at most 4 characters long)
 - ▶ Relationships between data (chromosomes have zero or more genes)

What is a Database?

- ▶ A data collection that needs to be :
 - ▶ Organized
 - ▶ Searchable
 - ▶ Up-to-date
- ▶ Challenge:
 - ▶ Change “meaningless” data into useful, accessible information

A spreadsheet can be a Database

A spreadsheet contains:

- ▶ Rectangular data
- ▶ Structured
- ▶ No metadata

Search tools:

- ▶ Excel
- ▶ grep
- ▶ python/R

SNP ID	SNPSeq ID	Gene	+primer	-primer	Hap A	Hap B	Hap C
D1Mit160_1	10.MMHAP67FLD1.seq	lymphocyte antigen 84	AAGGTAAAA GGCAATCAG CACAGCC	TCAACCTGG AGTCAGAGG CT	C	—	A
M-05554_1	12.MMHAP31FLD3.seq	procollagen, type III, alpha	TGCGCAGAA GCTGAAGTC TA	TTTTGAGGT GTTAATGGTT CT	C	—	A
M-05554_2	X60184	complement component factor i	ACTTCCAGC CCTGGCTCT	ATATGCCACC AAGAAGCA	A	C	—
M-09947_3	AF067835	caspase 8	TCACAGAGG GAAACATGA AG	CTCCACATTG AACCAGAGC A	G	C	T
M-11415_1	U02023	insulin-like growth factor binding protein	GGGAAAAGC CTGAAAGAA GC	AGCTGAAAC CGGACATCA AT	T	G	—
D1Mit284_3	J05234	nucleolin	TGTTGGAAC CGACTTCTTC A	AAGAGTCAA AGAAATTATG GAATGA	G	T	T

A filesystem can be a Database

Hierarchical data:

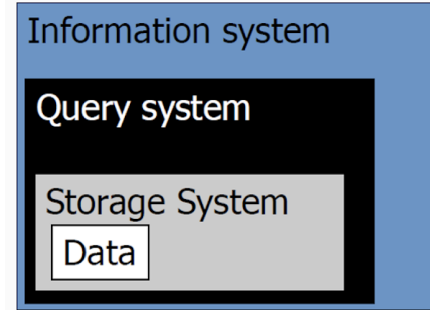
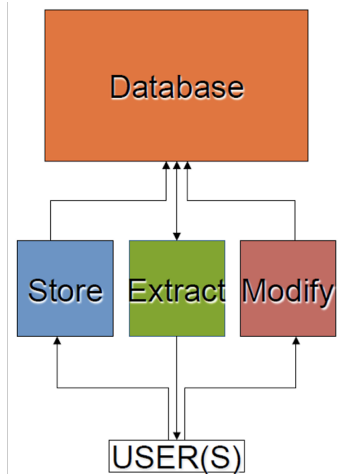
- ▶ Some metadata:
File, symlink, etc
- ▶ Unstructured

Search tools:

- ▶ ls
- ▶ find
- ▶ grep

```
--(/projectnb)-----
--(2019-02-21:16:27)-- tree bubhub | head -n 40
bubhub
├── bamcmp.simg
├── bubhub-conda
│   ├── contributors.txt
│   └── package_recipes
│       ├── args
│       │   ├── bld.bat
│       │   ├── build.sh
│       │   └── meta.yaml
│       ├── bash_kernel
│       ├── blast
│       ├── bx-python
│       │   ├── build.sh
│       │   ├── bx-python
│       │   └── build
│       │       ├── bdist.linux-x86_64
│       │       ├── lib.linux-x86_64-3.6
│       │       └── bx
│       │           ├── align
│       │           │   ├── axt.py
│       │           │   ├── _core.cpython-36m-x86_64-linux-gnu.so
│       │           │   ├── core.py
│       │           │   ├── _epo.cpython-36m-x86_64-linux-gnu.so
│       │           │   ├── epo.py
│       │           │   ├── epo_tests.py
│       │           │   ├── _init_.py
│       │           │   ├── lav.py
│       │           │   ├── lav_tests.py
│       │           │   ├── maf.py
│       │           │   ├── maf_tests.py
│       │           │   ├── score.py
│       │           │   ├── score_tests.py
│       │           │   ├── sitemask
│       │           │   │   ├── core.py
│       │           │   │   ├── _cpg.cpython-36m-x86_64-linux-gnu.so
│       │           │   │   ├── cpg.py
│       │           │   │   ├── _init_.py
│       │           │   │   ├── quality.py
│       │           │   │   └── sitemask_tests.py
│       │           └── tools
```

Organization and Types of Databases



Organization and Types of Databases

Every database has tools that: Store, Extract, Modify

Flat file databases (flat DBMS)

- Simple, restrictive, table

Hierarchical databases

- Simple, restrictive, tables

Relational databases (RDBMS)

- Complex, versatile, tables

Object-oriented databases (ODBMS)

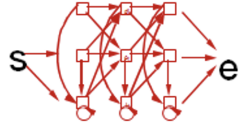
- Data warehouses and distributed databases

Unstructured databases (object store DBs)

Where do the data come from?

ctgccgatagc
MKLVDDYTR

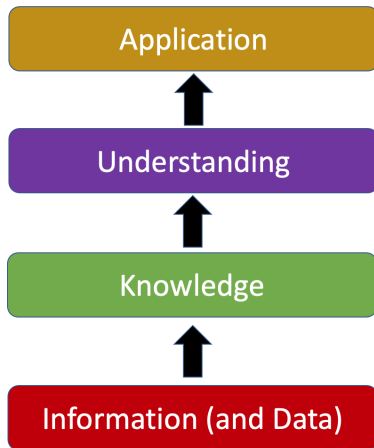
literature



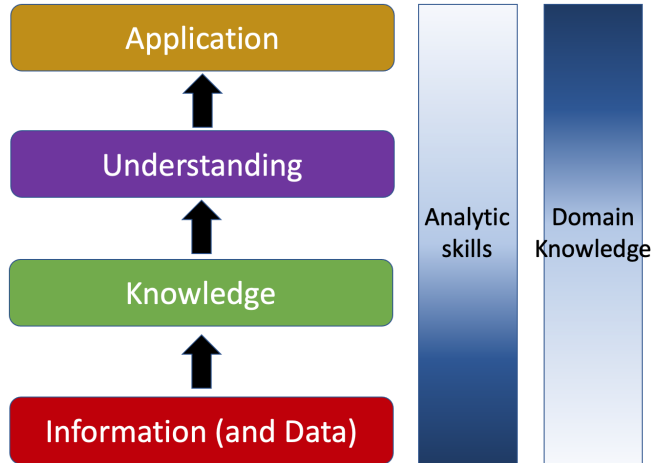
Information

New knowledge

Using biomedical data



Using biomedical data



Types of Biological Data

Primary data types (observed properties):

- ▶ Molecular Sequence: nucleic or amino acids
- ▶ Quantity: DNA, RNA, Protein, cell count, metabolites
- ▶ Locality: membrane, nucleus, epithelium
- ▶ Structure: 3D conformation, proximity, size

Secondary data types (inferred properties):

- ▶ Molecular Function
- ▶ Dynamics
- ▶ Relation: multiplicity, distribution, binding
- ▶ Association: co-occurrence, correlation
- ▶ Predicted: computational models

Types of Biological Databases

Primary Databases:

- ▶ Original submissions by experimentalists
- ▶ Content controlled by the submitter
- ▶ Examples: GenBank, Trace, SRA, SNP, GEO

Secondary databases:

- ▶ Results of analysis of primary databases
- ▶ Aggregate of many databases
- ▶ Content controlled by third party (NCBI)
- ▶ Examples: NCBI Protein, Refseq, RefSNP, GEO datasets, UniGene, Homologene, Structure, Conserved Domain

The NCBI hosts more than 50 different tools and databases

National Center for Biotechnology Information (NCBI):

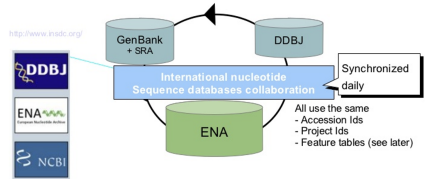
- ▶ **PubMed** - search for citations/papers
- ▶ **BLAST** - search for sequences
- ▶ **Nucleotide** - all nucleotide sequences (DNA, etc)
- ▶ **Genome** - published genomes
- ▶ **Protein** - amino acid sequences
- ▶ **SNP (dbSNP)** - all human SNPs
- ▶ **dbGAP** - Controlled access datasets

International Sequence Database Collaboration

Primary sequence dbs are synchronised and every sequence receives a unique identifier

All database maintainers assign and share a unique **accession number** (AC) to each sequence – besides their own ID number – (info at NCBI). Sequences can get updated, and the accession number is extended with a version number, e.g. .1 (see SVA)

Example of acc number: BC010109.2



[http://en.wikipedia.org/wiki/Accession_number_\(bioinformatics\)](http://en.wikipedia.org/wiki/Accession_number_(bioinformatics))

- ▶ International Sequence Database Collaboration
- ▶ National Centre for Biotechnology Information (NCBI)
- ▶ European Nucleotide Archive (ENA)
- ▶ DNA Data Bank of Japan (DDBJ)

PubMed is a search engine for literature

- ▶ Citation/publication databases
- ▶ Medline: <https://www.nlm.nih.gov/bsd/pmresources.html>
 - ▶ NLM journal citation database.
 - ▶ Includes citations 5,600 scholarly journals
- ▶ PubMed <https://www.ncbi.nlm.nih.gov/pubmed/>
 - ▶ Includes MEDLINE
 - ▶ journals/manuscripts deposited in PMC
 - ▶ NCBI Bookshelf

Searching PubMed with MeSH terms

MeSH (Medical Subject Headings) is the NLM controlled vocabulary used for indexing articles for PubMed.

- ▶ the U.S. National Library of Medicine's controlled vocabulary
- ▶ arranged in a hierarchical manner called the MeSH Tree Structures updated annually



The screenshot shows the PubMed website interface. At the top, there is a navigation bar with "NCBI", "Resources", and "How To" links. The main search bar contains the query "(microRNA[Title]) AND bastola[Author]" and a "Search" button. Below the search bar, the results are displayed in a list format. The first result is titled "Contribution of bioinformatics prediction in microRNA-based cancer therapeutics." and is authored by Banwait JK¹ and Bastola DR². The abstract text is visible below the title. On the right side of the page, there are links to "Full text links" and "Save items".

NCBI Resources How To Sign in to NCBI

PubMed.gov US National Library of Medicine National Institutes of Health

PubMed (microRNA[Title]) AND bastola[Author] Search

Create RSS Create alert Advanced Help

Format: Abstract

Send to

Full text links

ELSEVIER FULL-TEXT ARTICLE

PMC Full text

Save items

Add to Favorites

Ady Drug Deliv Rev. 2015 Jan;81:94-103. doi: 10.1016/j.addr.2014.10.030. Epub 2014 Nov 6.

Contribution of bioinformatics prediction in microRNA-based cancer therapeutics.

Banwait JK¹, Bastola DR².

Author information

Abstract

Despite enormous efforts, cancer remains one of the most lethal diseases in the world. With the advancement of high throughput technologies, massive amounts of cancer data can be processed and analyzed. Bioinformatics provides a platform to assist biologists in

Center for Data Science

Google Scholar

Google Scholar is another alternative for finding publications:



W. Evan Johnson 

Professor of Medicine
Verified email at rutgers.edu

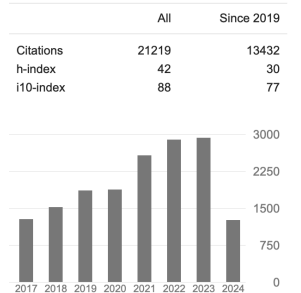
 FOLLOW

[Data Science](#) [Computational Biology](#) [Bioinformatics](#) [Metagenomics](#) [Tuberculosis](#)

☐ TITLE  	CITED BY	YEAR
☐ Adjusting batch effects in microarray expression data using empirical Bayes methods WE Johnson, C Li, A Rabinovic Biostatistics 8 (1), 118-127	7202	2007
☐ The sva package for removing batch effects and other unwanted variation in high-throughput experiments JT Leek, WE Johnson, HS Parker, AE Jaffe, JD Storey Bioinformatics 28 (6), 882-883	4502	2012
☐ Tackling the widespread and critical impact of batch effects in high-throughput data JT Leek, RB Scharpf, HC Bravo, D Simcha, B Langmead, WE Johnson, ... Nature Reviews Genetics 11 (10), 733-739	2091	2010

Cited by

[VIEW ALL](#)



Public access

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0 articles

97 articles

Collections of DNA/RNA sequences

GenBank is an annotated collection of all publicly available DNA sequences

GenBank: <https://www.ncbi.nlm.nih.gov/genbank/>

- ▶ Flat file
- ▶ DNA only sequence database
- ▶ Archival in nature: Historical, Redundant
- ▶ Sample GenBank record (accession number U49845)
 - ▶ NCBI: <https://www.ncbi.nlm.nih.gov/nuccore/U49845>
 - ▶ ENA: <https://www.ebi.ac.uk/ena/data/view/U49845>
 - ▶ DDBJ: <http://getentry.ddbj.nig.ac.jp/top-e.html>



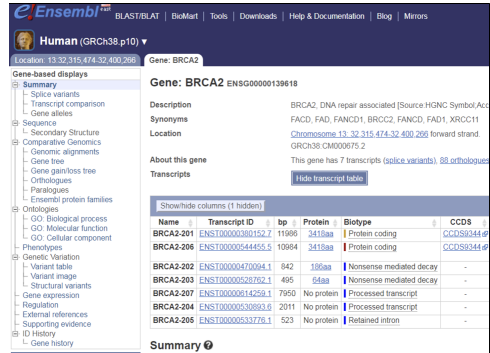
Header

- Title
- Taxonomy
- Citation

Features (AA seq)

Ensembl (European Bioinformatics Institute)

- ▶ Comprehensive DNA/RNA sequence and annotation database
- ▶ Ensembl annotate genes, computes multiple alignments, predicts regulatory function and collects disease data
- ▶ Analysis tools:
 - ▶ BLAST
 - ▶ BioMart
 - ▶ Variant Effect Predictor



The screenshot shows the Ensembl genome browser interface for the Human (GRCh38.p10) genome. The gene of interest is BRCA2, located on Chromosome 13 at position 32,315,474-32,400,266. The page displays various annotations including splice variants, transcript comparisons, and gene alleles. A table of transcripts is shown, listing transcript IDs, lengths, protein lengths, biotypes, and CCDS IDs.

Name	Transcript ID	bp	Protein	Biotype	CCDS
BRCA2-201	ENST00000380152.7	11986	3418aa	Protein coding	CCDS9344.6
BRCA2-206	ENST00000544455.5	10984	3418aa	Protein coding	CCDS9344.6
BRCA2-202	ENST00000470094.1	842	186aa	Nonsense mediated decay	-
BRCA2-203	ENST00000528762.1	495	64aa	Nonsense mediated decay	-
BRCA2-207	ENST00000614259.1	7950	No protein	Processed transcript	-
BRCA2-204	ENST00000530893.6	2011	No protein	Processed transcript	-
BRCA2-205	ENST00000533776.1	523	No protein	Retained intron	-

Databases for reference sequences

Reference genomes

- ▶ NCBI Genome -
<https://www.ncbi.nlm.nih.gov/datasets/genome/>
 - ▶ Contains both Genbank and Refseq accessions
- ▶ Ensembl - <https://useast.ensembl.org/index.html>

microRNAs

- ▶ miRbase: <https://mirbase.org> allows you to search and browse miRNAs from a host of species
- ▶ Enables download of genomic coordinates for miRNAs (.gff) and sequence (.fa)

rRNAs and tRNAs databases

- ▶ rRNAs
- ▶ Greengenes - <https://greengenes.secondgenome.com>
- ▶ Silva - <https://www.arb-silva.de>
- ▶ tRNAs
 - ▶ gtRNAdb- <http://gtrnadb.ucsc.edu>

Species-specific databases

For other well studied model organisms, you can often find collections of resources and tools:

- ▶ Flybase - *D. melanogaster*
- ▶ Wormbase - *C. elegans*
- ▶ Zfin - *D. rerio*
- ▶ Influenza (link!)

RNACentral



- ▶ Developed by EBI and Wellcome Trust
- ▶ Integration of many other up-to-date RNA resources and tools

[Click here for an overview of core functions](#)

Existing datasets – sequencing data (SRA or GEO)

dbGaP: genotype-phenotype interactions in humans

- ▶ Most studies contain PHI and are subject to strict access control and usage regulations
- ▶ Applications required to be granted access to datasets
- ▶ Any data with sensitive health information must be stored, handled, and interacted with according to appropriate regulations

SRA - Sequencing Read Archive

- ▶ Raw sequencing data and alignment info
- ▶ Metagenomics, environmental samples, biomedical sequencing

Download via:

- ▶ SRA-toolkit
- ▶ FTP links on EMBL ENA

A quick demonstration of SRA-toolkit and EMBL-ENA

- ▶ SRA-toolkit is the official tool released by NCBI to directly download SRA files
 - ▶ Notorious for being obtuse to use and confusing commands / documentation
- ▶ EMBL-ENA hosts FTP links directly
 - ▶ Most but not all SRA accessions available
 - ▶ Have to use wget, curl, or other methods to download

A quick demonstration of SRA-toolkit

Download a file for an asthma host microbiome dataset

```
## attach the sratoolkit  
module load sratoolkit  
  
## Save accession to download  
acc="SRR1528344"  
  
## Download using fastq-dump  
fastq-dump $acc  
# option --split-3 is needed for paired end reads  
  
## don't forget to compress the file!  
gzip $acc.fastq
```


A quick demonstration of SRA-toolkit

Download all files for an asthma host microbiome dataset

```
accs=( $( cat SRR_Acc_List.txt ) )  
for i in $(seq 0 ${#accs[@]})  
do  
    fastq-dump ${accs[i]};  
    gzip ${accs[i]}*  
done
```

Batch script (SLURM)

```
#!/bin/bash
#SBATCH --job-name=microbiome_download
#SBATCH --mem=1G
#SBATCH --time=01:00:00

module load sratoolkit
cd $HOME/tmp/

accs=( $( cat SRR_Acc_List.txt ) )
for i in $(seq 0 ${#accs[@]})
do
    fastq-dump ${accs[i]};
    gzip ${accs[i]}*
done
```

Save as a file and use sbatch to submit

Batch array (SLURM)

```
#!/bin/bash
#SBATCH --job-name=microbiome_download
#SBATCH --output=asthma.out
#SBATCH --array=0-27
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G
#SBATCH --time=00:20:00

module load sratoolkit
cd /scratch/$USER

accs=( $( cat SRR_Acc_List.txt ) )
acc_number=${accs[$SLURM_ARRAY_TASK_ID]}

fastq-dump --gzip $acc_number
```

Save as a file and use sbatch to submit

Gene Expression Data

GEO - Genome Expression Omnibus

- ▶ Tied to SRA via Bioproject ID
- ▶ GEO also contains processed data, typically specific to a publication
- ▶ You may find: Alignments (BAM/SAM), visualizations (.bg, .bed) and intermediate results (delimited formats)

GEOquery example

```
# Load the geoquery library: BiocManager::install("GEOquery")
library(GEOquery)

# Search for a dataset in GEO
geo_search <- getGEO("GSE1297", GSEMatrix = TRUE)

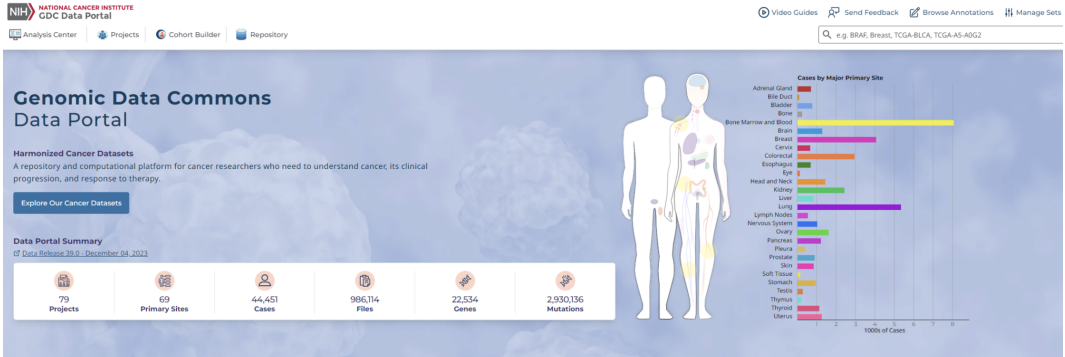
# Display basic information about the dataset
print(geo_search)

# Extract expression data
expression_data <- exprs(geo_search[[1]])

# Display the first few rows of expression data
head(expression_data)
```

Domain-specific data

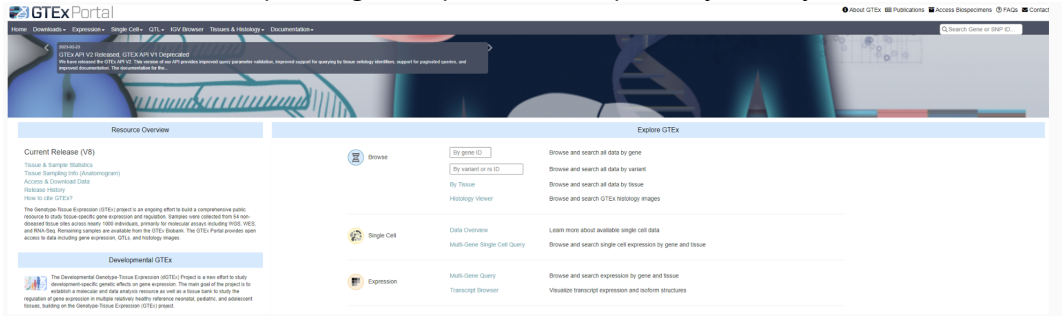
TCGA / NCI Genomic Data Commons



<https://portal.gdc.cancer.gov>

Tissue-specific gene expression

GTEx collects tissue-specific gene expression from primarily healthy individuals



The screenshot shows the GTEx Portal homepage. At the top, there is a navigation bar with links: Home, Downloads, Expression, Single Cell, QTL, KIV Browser, Tissues & Histology, and Documentation. A search bar is located on the right. Below the navigation bar, there is a large banner image with a DNA double helix and a bar chart. A message box states: "GTEx API V2 Released, GTEx API V1 Deprecated. We have released the GTEx API V2. This version of our API provides improved query parameter validation, improved support for querying by tissue/histology identifier, support for paginated content, and improved documentation. The documentation for the..."

The main content area is divided into two columns. The left column is titled "Resource Overview" and includes links for "Current Release (V8)", "Tissue & Sample Statistics", "Tissue Sampling Info (Anatomogram)", "Access & Download Data", "Release History", and "How to cite GTEx?". Below these links is a paragraph describing the GTEx project: "The Genotype Tissue Expression (GTEx) project is an ongoing effort to build a comprehensive public resource to study tissue-specific gene expression and regulation. Samples were collected from 54 non-diseased tissue sites across nearly 1000 individuals, primarily for molecular assays including microarrays and RNA-Seq. Remaining samples are available from the GTEx Biobank. The GTEx Portal provides open access to data including gene expression, QTLs, and histology images."

The right column is titled "Explore GTEx" and contains a table of links:

Icon	Link	Description
Browse	By gene ID	Browse and search all data by gene
	By variant or rs ID	Browse and search all data by variant
	By Tissue	Browse and search all data by tissue
	Histology Viewer	Browse and search GTEx Histology images
Single Cell	Data Overview	Learn more about available single cell data
	Multi-Gene Single Cell Query	Browse and search single cell expression by gene and tissue
Expression	Multi-Gene Query	Browse and search expression by gene and tissue
	Transcript Browser	Visualize transcript expression and isoform structures

<https://www.gtexportal.org/home/>

DNA regulatory elements

ENCODE contains a host of information about DNA regulatory elements



<https://www.encodeproject.org>

Protein databases

NCBI Protein

Protein sequence database: <https://www.ncbi.nlm.nih.gov/protein/>

NCBI
Resources
How To
Sign in to NCBI

Protein
Protein
BRAC
Search
Create alert
Advanced
Help

Species
Animals (79,941)
Plants (14,521)
Fungi (12,247)
Protists (4,931)
Bacteria (7,704)
Archaea (34)
Viruses (571)
Customize ...

Source databases
PDB (232)
RefSeq (73,065)
UniProtKB / Swiss-Prot (516)
Customize ...

Genetic compartments
Chloroplast (2)
Mitochondrion (13)
Plasmid (12)
Plastid (2)

Sequence length
Custom range ...

Summary
20 per page
Sort by Default order
Send to
Filters: Manage Filters

See [braC branched-chain amino acid ABC transporter substrate-binding protein BraC](#) in the Gene database
brac reference sequences [Protein \(1\)](#)

See the [results of this search \(87 items\)](#) in our new [Identical Protein Groups](#) database.

Items: 1 to 20 of 120705

1. [BRAC_{partial} \[Poeciliopsis prolifica\]](#)
106 aa protein
Accession: JAO55668.1 GI: 958322777
[BioProject](#) [Nucleotide](#) [Taxonomy](#)
[GenPept](#) [Identical Proteins](#) [FASTA](#) [Graphics](#)

2. [BRAC \[Pseudomonas putida BIRD-1\]](#)
371 aa protein

Results by taxon
Top Organisms [Tree](#)
[Homo sapiens \(1503\)](#)
[Plasmodium falciparum \(1448\)](#)
[Mycobacterium abscessus \(1298\)](#)
[Mus musculus \(1240\)](#)
[Rhizophagus irregularis \(1203\)](#)
[All other taxa \(114013\)](#)
More...

Find related data
Database: Select
Find items

Search details
BRAC[All Fields]

Center for Data Science

Genpept

Chain A, Structure Of The Btb (tramtrack And Bric A Gigaxonin

PDB: 2PPI_A

[Identical Proteins](#) [FASTA](#) [Graphics](#)

[Go to:](#) ☐

LOCUS 2PPI_A 144 aa linear PRI 26-OCT-2017

DEFINITION Chain A, Structure Of The Btb (tramtrack And Bric A Brac) Domain Human Gigaxonin.

ACCESSION 2PPI_A

VERSION 2PPI_A

DBSOURCE pdb: molecule 2PPI, chain 65, release Oct 22, 2017;
deposition: Apr 30, 2007;
class: Structural Protein;
source: Mmdb_id: [46639](#), Pdb_id 1: 2PPI;
Exp. method: X-Ray Diffraction.

KEYWORDS .

SOURCE Homo sapiens (human)

ORGANISM [Homo sapiens](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
Catarrhini; Hominidae; Homo.

REFERENCE 1 (residues 1 to 144)

AUTHORS Amos,A., Turnbull,A.P., Tickle,J., Keates,T., Bullock,A., S.P.,
Burgess-Brown,N., Debreczeni,J.E., Ugochukwu,E., Umeano,C.,
Pike,A.C.W., Papagrigoriou,E., Sundstrom,M., Arrowsmi,C.H.,
Weigelt,J., Edwards,A., Von Delft,F. and Knapp,S.

TITLE Structure Of The Btb (Tramtrack And Bric A Brac) Do Human Gigaxonin

JOURNAL Unpublished

[Region](#)

[Region](#)

[SecStr](#)

[SecStr](#)

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[SecStr](#)

[SecStr](#)

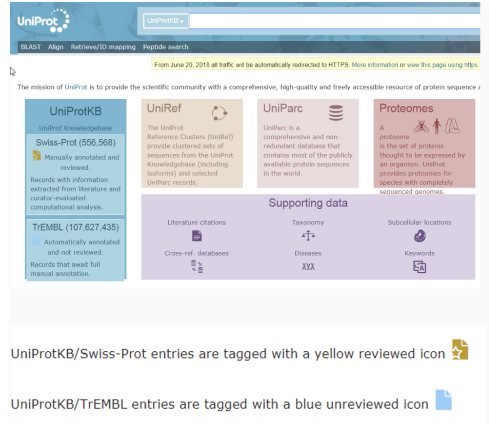
```

//region_name= BTB
/note="BTB/POZ domain; pfam00651"
/db_xref="CDD:279045"
46..96
/region_name="Domain 2"
/note="NCBI Domains"
49..144
/region_name="BTB"
/note="Broad-Complex, Tramtrack and Bric a brac;
smart00225"
/db_xref="CDD:197585"
49..55
/sec_str_type="sheet"
/note="strand 1"
56..62
/sec_str_type="sheet"
/note="strand 2"
63..70
/sec_str_type="helix"
/note="helix 2"
71..79
/sec_str_type="helix"
/note="helix 3"
89..94
/sec_str_type="sheet"
/note="strand 3"
98..109
/sec_str_type="helix"
/note="helix 4"
120..130
/sec_str_type="helix"
/note="helix 5"

```

Uniprot

- ▶ The Universal Protein Resource
- ▶ Comprehensive resource for protein sequence and annotation data
- ▶ Collaboration between:
 - ▶ EMBL-EBI
 - ▶ Swiss Institute of Bioinformatics
 - ▶ Protein Information Resource
- ▶ Entries in two categories:
 - ▶ Swiss-Prot (experimentally verified)
 - ▶ TrEMBL (computer-annotated)
- ▶ <http://www.uniprot.org/>



The screenshot shows the UniProt website interface. At the top, there is a navigation bar with the UniProt logo and a search bar. Below the navigation bar, a banner states: "From June 20, 2018 all traffic will be automatically redirected to HTTPS. More information or view this page using https". The main content area features four columns of information:

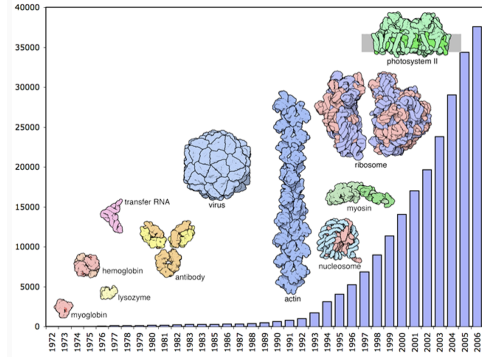
- UniProtKB**: UniProt Knowledgebase. It lists two categories: Swiss-Prot (556,568) with a yellow reviewed icon and TrEMBL (107,627,435) with a blue unreviewed icon. Descriptions for each category are provided.
- UniRef**: The UniProt Reference Clusters (UniRef) provide clustered sets of sequences from the UniProt Knowledgebase (including isoforms) and selected UniParc records.
- UniParc**: UniParc is a comprehensive and non-redundant database that contains most of the publicly available protein sequences in the world.
- Proteomes**: A proteome is the set of proteins thought to be expressed by an organism. UniProt provides proteomes for species with completely sequenced genomes.

Below these columns is a section titled "Supporting data" which includes links to Literature citations, Cross-ref. databases, Taxonomy, Diseases, and Subcellular locations. At the bottom, two explanatory text blocks are shown:

- UniProtKB/Swiss-Prot entries are tagged with a yellow reviewed icon 
- UniProtKB/TrEMBL entries are tagged with a blue unreviewed icon 

Protein Structure database - PDB

- ▶ Protein Data Bank (PDB)
<http://www.rcsb.org/>
- ▶ Dedicated to 3D structure of proteins and peptides
- ▶ ~150,000 predicted and experimental (solved) structures



PDB: kinesin 6



149174 Biological Macromolecular Structures Enabling Breakthroughs in Research and Education

[Advanced Search](#) | [Browse by Annotations](#)





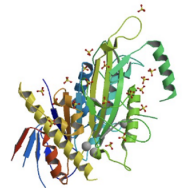







[Structure Summary](#)
[3D View](#)
[Annotations](#)
[Sequence](#)
[Sequence Similarity](#)
[Structure Similarity](#)
[Experiment](#)

Biological Assembly 1 ?



5X3E

kinesin 6

DOI: [10.2210/pdb5X3E/pdb](https://doi.org/10.2210/pdb5X3E/pdb)

Classification: [MOTOR PROTEIN](#)

Organism(s): [Caenorhabditis elegans](#)

Expression System: [Escherichia coli-Thermus thermophilus shuttle vector pTRH17](#)

Deposited: 2017-02-04 Released: 2017-04-19

Deposition Author(s): [Chen, Z.](#), [Guan, R.](#), [Zhang, L.](#)

Funding Organization(s): Chinese Key Research Plan-Protein Sciences; National Natural Science Foundation of China; Junior One Thousand Talents program; National Science Foundation of China; 863 Program

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.61 Å

R-Value Free: 0.240

R-Value Work: 0.209

wwPDB Validation

Metric	Percentile Ranks	Value
Rfree		0.240
Clashscore		7
Ramachandran outliers		0
Sidechain outliers		1.1%
RSRZ outliers		13.0%

3D View: [Structure](#) | [Electron Density](#) | [Ligand Interaction](#)

Standalone Viewers
[Protein Workshop](#) | [Ligand Explorer](#)

Protein Family Database

- ▶ <http://pfam.xfam.org/family/piwi>
- ▶ Pfam is a database of protein families that includes their annotations and multiple sequence alignments generated using Hidden Markov Models

Family: *Piwi* (PF02171)

 139 architectures
  3730 sequences
  4 interactions
  568 species
  103 structures

Summary

Domain organisation

Clan
Alignments
HMM logo
Trees
Curation & model
Species
Interactions
Structures

Jump to...

Domain organisation

Below is a listing of the unique domain organisations or architectures in which this domain is found. [More...](#)

There are 893 sequences with the following architecture: ArgoN, Argol1, PAZ, Argol2, ArgoMid, Piwi
[X1WG39_DANRE](#) [Danio rerio (Zebrafish)] (Brachydanio rerio)] Uncharacterized protein {ECO:0000313|Ensembl:ENSDARP00000129194} (858 residues)

[Show](#) all sequences with this architecture.

There are 678 sequences with the following architecture: Piwi
[Z4YLE4_MOUSE](#) [Mus musculus (Mouse)] Piwi-like protein 4 {ECO:0000313|Ensembl:ENSMUSP00000111307} (458 residues)

[Show](#) all sequences with this architecture.

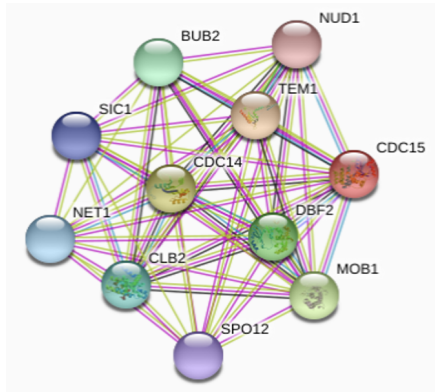
There are 581 sequences with the following architecture: ArgoN, Argol1, PAZ, Argol2, Piwi
[J3NVY6_GAGT3](#) [Gaeumannomyces graminis var. tritici (strain R3-111a-1) (Wheat and barley take-all root rot fungus)] Uncharacterized protein {ECO:0000313|EMBL:EJT75516.1, ECO:0000313|EnsemblFungi:GGTG_05449T0} (1022 residues)

[Show](#) all sequences with this architecture.

There are 447 sequences with the following architecture: PAZ, Piwi
[V4B7N4_LOTGI](#) [Lottia gigantea (Giant owl limpet)] Uncharacterized protein {ECO:0000313|EMBL:ES084639.1} (791 residues)


Protein-Protein Interaction Database

- ▶ STRING: <https://string-db.org/>
- ▶ Search Tool for the Retrieval of Interacting Genes/Proteins
- ▶ Database of protein/protein interactions
- ▶ Information from numerous sources:
 - ▶ experimental data
 - ▶ computational prediction methods
 - ▶ public text collections
- ▶ Expressed as interaction graphs:
 - ▶ Nodes: Network nodes represent proteins
 - ▶ Edges: Edges represent protein-protein associations



Curated resources

Data vs Annotation Database

- ▶ RefSeq: curated nonredundant biological sequences
<https://www.ncbi.nlm.nih.gov/refseq/>
 - ▶ Source: Genbank (INSDC)
 - ▶ Annotated: Community collaboration, automated computer, NCBI staff curation
- ▶ Advantages of using RefSeq
 - ▶ Non-redundancy
 - ▶ Curated, validated
 - ▶ Format consistency
 - ▶ Distinct accession series
 - ▶ Updates to reflect current sequence data and biology

RefSeq Annotations

mRNAs and Proteins

NM_123456

Curated mRNA

NP_123456

Curated Protein

NR_123456

Curated non-coding RNA

XM_123456

Predicted mRNA

XP_123456

Predicted Protein

XR_123456

Predicted non-coding RNA

Gene Records

NG_123456

Reference Genomic Sequence

Chromosome

NC_123455

Microbial replicons, organelle

AC_123455

Alternate assemblies

Assemblies

NT_123456

Contig

NW_123456

WGS Supercontig

Avoid using outdated or abandoned tools / databases

Keep in mind the following:

- ▶ Check when it was last updated (ideally recently)
- ▶ Check for an associated peer-reviewed publication
- ▶ Check for a github repo
 - ▶ How recent was the last commit
 - ▶ Are they responsive to issues?

Session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.5
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.0
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.4.2    fastmap_1.2.0     cli_3.6.5         tools_4.4.2
## [5] htmltools_0.5.8.1 rstudioapi_0.17.1 yaml_2.3.10       rmarkdown_2.29
## [9] knitr_1.50        xfun_0.52         digest_0.6.37     rlang_1.1.6
## [13] evaluate_1.0.3
```